2026Year 2nd Semester Syllabus

Created Date	2026-08-07 08:37:05	Last-Modified	2026-08-07 16:36:42
Course Title	QUANTUM COMPUTER PROGRAMMING	Course Code-Section	CAS3140-01
Credit/Time/ Experiment,Lab,Practical Technique Time	3/Mon5,6,Wed6	Department	School of Computing
Time	Mon5,6,Wed6	Location	EngHD408
Exam Date & Time	Midterm exam	Final exam	
Class Language	English	Evaluation Type	Absolute evaluation

Instructor's Profile	Name	Burgstaller bernd	Contact Information	Telephone	02-2123-5728
	Department	DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING		Mail	BBURG@YONSEI.AC.KR
	Office	Engineering Hall 4, D910		Interview information	Zoom consultations upon appointment request via email

TA's Name & Contact Information	Name	To be announced in the first class	Contact Information	Telephone	

Course Description Brief Introduction of the Course	Quantum computers have the potential to solve difficult decision-making and optimization problems for which no efficient classical algorithms exist. Writing quantum algorithms is radically different from programming classical computers and requires an understanding of quantum principles and their mathematical foundations. This course approaches quantum computing from a computer-science perspective. It introduces the principles of quantum mechanics as they relate to information (quantum states) and computation (quantum gates and circuits). The computational advantages of quantum information over classical computers, i.e., superposition, entanglement, and interference, are demonstrated with principal quantum algorithms, including quantum key distribution, quantum query algorithms, integer factorization, and search. The course covers classical control of quantum computations and touches upon error mitigation techniques for noisy intermediate-scale (NISQ) quantum devices. Course participants will gain practical experience in programming quantum computers on the IBM Quantum Platform and in simulating quantum circuits on their notebook computer.			
Course Goals	1.	Korean	모든 양자 계산의 핵심이 되는 큐비트의 중첩, 얽힘, 간섭에 관한 양자역학적 기초를 확실히 이해한다.	25%
		English	Acquire a firm understanding of the quantum-mechanical foundations of qubit superposition, entanglement, and interference at the heart of all quantum computations.	
	2.	Korean	양자 키 분배, 양자 텔레포테이션, 도이치 문제, 번스타인-바지라니 알고리즘, 양자 고속 푸리에 변환(Quantum FFT)과 같은 초기 양자 알고리즘을 이해하고, 이를 코딩하여 양자 처리 장치(QPU)에서 실행할 수 있다.	25%
		English	Understand the early quantum algorithms such as Quantum Key Distribution, Quantum Teleportation, Deutsch's Problem, Bernstein-Vazirani, and Quantum FFT, and be able to code and execute them on a QPU.	
	3.	Korean	해밀토니안 동역학의 양자 시뮬레이션과 같은 최근의 근시일 양자 알고리즘을 이해한다.	25%
		English	Know recent near-term quantum algorithms like the quantum simulation of Hamiltonian dynamics.	

	4.	Korean	근시일 양자 워크로드를 양자 처리 장치(QPU)에서 실행하기 위한 양자 회로 컴파일 파이프라인과 오류 완화 기법을 원리적으로 이해하고 제어할 수 있다.	25%
		English	Understand and control, in principle, the quantum circuit compilation pipeline and error mitigation techniques to execute near-term quantum workloads on QPUs.	
	5.	Korean		0%
		English		

Core Competencies	The total measurable competencies must be 100%. Each course objective should set the competency as 25%. The core and major competencies should equal at least 50%.					
	논리적사고	50%	프로그래밍능력	30%	문제해결능력	20%
Sub-Competencies/Learning Unit1						
Sub-Competencies/Learning Unit2						
Sub-Competencies/Learning Unit3						
Core Competencies(Liberal Arts)Major competency(	**Must reflect the interrelationship between core competencies (elective courses) and major competencies (major studies).**					
논리적사고						
Sustainable Development Goals						
Average Recommended Amount of Learning per	Average Reading Volume		Average amount of writing(Based on A4)			

Course Methods (%) Total Amount 100	Lecture	Practice Training	Presentation	Dabate	Team Project
	80%	20%	0%	0%	0%
Course Methods 2 Select Relevant Items	PBL Subject	Capstone Design	CBL, Social Innovation Course	Flipped Classroom	Work Experience,Internsh

Grading Policy(%) Total Amount 100 Free Input for Other Information	Midterm exam	Final exam	Quiz	Individual Assignment	Team Assignment	Attendance	Others
	32%	32%	0%	32%	0%	0%	4%

Assignment/ Report, Project Guide	Title of Assignment/Project Name, and Method of Filling Out	Submission Deadline	Type of Submission and Method
Prerequisite	- An introductory programming course (preferably on Python), e.g., CAS1100-01 Computer Programming. - A course in linear algebra, e.g., MAT-2011 Linear Algebra and Its Applications. - A course on algorithms and data structures, e.g., CAS2103-01 Data Structure.	Online Course Address	

Course Material	Course Material Name	Author	Publisher	Publish Year	ISBN
			Online, available at		

참고자료	A Course in Quantum Computing, Volume 1	Michael Loceff	https://lapastillaroja.net/wp-content/uploads/2016/09/Intro_to_QC_Vol_1_Loceff.pdf	2015	n/a
참고자료	Quantum Computing for Computer Scientists	Noson S. Yanofsky and Mirco A. Mannucci	Cambridge University Press	2008	978-0-521-87996-5
참고자료	Quantum Computation and Quantum Information, 10th aniversary edition	Michael A. Nielsen & Isaac L. Chuang	Cambridge University Press	2010	978-1-107-00217-3

Main Learner Precautions	Course format: - The course contents will be delivered as a series of face-to-face lectures. - Along with the lectures, small problem sets (assignments) will be issued for course participants to work on practical examples of the concepts studied in the lectures (algebra and coding). All assignments are individual assignments, to be solved at home, typically within a one-week timeframe (due dates will be clearly stated with each assignment). - Coding portions of assignments will involve code skeletons delivered in the form of Jupyter notebooks, that course participants can extend and execute on their local computer, using the VS Code editor. Our implementation language will be Python 3, with the Qiskit SDK. - The course language is English, including lectures, lab assignments, and exams. AI policy: - Generative AI is permissible as a supplementary tool for web search with the assignments, but not for solving the assignment questions themselves. - Exams will be conducted in-class, with paper and pencil. The use of AI tools with the exams is not allowed.
Attatchment	

Weekly Plan

week	Period	Weekly Topic & Contents	Remarks
1	2026-09-01 2026-09-07	Quantum information science, photons as qubits, the double-slit experiment.	(9.1.) Fall semester classes begin (9.3. - 9.7.) Course add and drop period
2	2026-09-08 2026-09-14	Hilbert space, single qubits, Bloch sphere representation of qubits.	
3	2026-09-15 2026-09-21	Unitary operations, single-qubit quantum gates, introduction to quantum circuits, Hermitian operators.	
4	2026-09-22 2026-09-28	Observables and measurements, measurement uncertainty, quantum hello-world, quantum key distribution protocol.	(9.24. - 9.25.) Chuseok Holidays 09.24 추석, 09.25 추석, 09.26 추석
5	2026-09-29 2026-10-05	The experimental basis of quantum computing and the quantum postulates.	(10.5.) Alternative Holiday of the National foundation Day 10.03 개천절, 10.05 대체공휴일(개천절)
6	2026-10-06 2026-10-12	Tensors, bi-partite Hilbert space, two-qubit gates, entanglement and Bell states, quantum parallelism, interference.	(10.7.) First third of the semester ends (10.9.) Hangul Proclamation Day 10.09 한글날
7	2026-10-13 2026-10-19	The no-cloning theorem, superdense coding.	(10.13. - 10.15.) Course withdrawal period

8	2026-10-20 2026-10-26	Midterm exam.	(10.20. - 10.26.) Midterm Examinations
9	2026-10-27 2026-11-02	Quantum teleportation, Deutsch’s algorithm.	(10.27. - 10.29.) Application Period for S/U evaluation
10	2026-11-03 2026-11-09	Bernstein-Vazirani’s algorithm, Simon's problem.	
11	2026-11-10 2026-11-16	Quantum FFT.	(11.13.) Second third of the semester ends
12	2026-11-17 2026-11-23	Shor's algorithm (factoring), Grover's algorithm (quantum search).	
13	2026-11-24 2026-11-30	Error suppression and mitigation on current NISQ devices.	
14	2026-12-01 2026-12-07	Qiskit quantum circuit transpilation pipeline.	
15	2026-12-08 2026-12-14	Self-study week.	(12.8. - 12.14.) Self-study
16	2026-12-15 2026-12-21	Final exam week.	(12.15. - 12.21.) Final Examinations

Attendance Requirement

• Students who are absent for more than one-third of the total class hours will receive an F or NP grade regardless of examination results.
• Courses without midterm and/or final examinations will hold regular classes during the examination periods.

Support for Students with Disabilities(SWDs)

• SWDs can request accommodations related to lectures, assignments, or tests by contacting the course professor at the beginning of semester. However, accommodations may vary depending on the essentiality of lecture and discretion of professors.

[Lecture]
- Visual Impairment: alternative, braille, enlarged reading materials, note-taker
- Physical Impairment: alternative reading materials, access to classroom, note-taker, assigned seat
- Hearing Impairment: note-taker/stenographer, recording lecture
- Intellectual Disability/Autism: note-taker

[Assignments and Test]
- Visual/Physical/Hearing Impairment: (reasonable) extra days for submission, alternative type of assignment, extended test time, alternative type of test, arranging separate test room, and proctors, test ghostwriter
- Intellectual Disability/Autism: (reasonable) extra days for submission, alternative type of assignment

Safety Precautions

• Students required to complete laboratory safety training must finish the online training before the semester begins and submit the certificate at the first class session. Failure to do so may result in restricted class participation.
• Students in physical activity courses should complete appropriate warm-up exercises before class. Those with cardiovascular or chronic respiratory conditions are advised to consult a physician before participating.

Course Administration

• Notices regarding class cancellations, make-up classes, classroom changes, and other course-related matters may be posted on LearnUs. For updates, please check LearnUs regularly.